SPACE & ASTRONOMY · GENERAL KNOWLEDGE

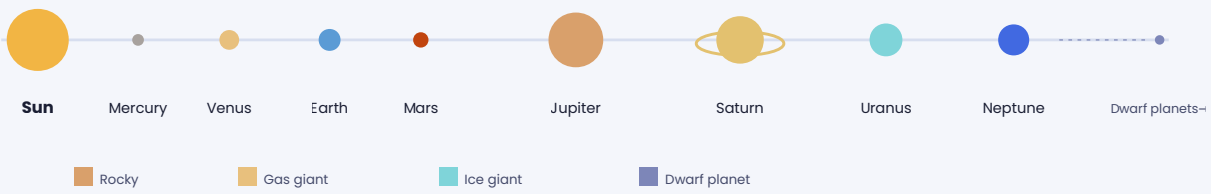
Cosmic Almanac

Planets, moons, stars, galaxies, black holes and the universe itself — with distances given in light-time so they're easy to remember and compare, not just memorise.

01

The Solar System

The Sun and everything bound to it by gravity: 8 planets, 5+ recognised dwarf planets, hundreds of moons, and countless comets and asteroids.



The eight planets in order, coloured by type — sizes and spacing are schematic, not to scale.

KEY POINTS

- Order from the Sun: **Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune.** (Mnemonic: *My Very Educated Mother Just Served Us Noodles.*)
- **Rocky / terrestrial planets** — Mercury, Venus, Earth, Mars: small, dense, solid surfaces.
- **Gas giants** — Jupiter, Saturn: mostly hydrogen and helium, no solid surface, by far the largest planets.
- **Ice giants** — Uranus, Neptune: mostly water, ammonia and methane ices around a small rocky core.
- **Dwarf planets** — Pluto, Eris, Haumea, Makemake, Ceres: round and orbit the Sun, but haven't cleared their orbital neighbourhood of other debris (this is why Pluto was reclassified in 2006).
- The Solar System's outer edge, the **Oort Cloud**, is thought to reach roughly 1–2 light-years from the Sun — about halfway to the nearest star.

02

Planets & Their Moons

Every planet's type, its key named moons, and one fact worth remembering.

Mercury ROCKY

Moons: none. Smallest planet; a year lasts 88 Earth days, but one full day-night cycle takes 176 Earth days.

Venus ROCKY

Moons: none. Hottest planet (~465°C, thick CO_2 atmosphere); spins backwards and slower than it orbits the Sun.

Earth ROCKY

Moons: 1 — the Moon (Luna). The only planet known to host life.

Mars ROCKY

Moons: 2 — Phobos and Deimos (small, potato-shaped). Home to Olympus Mons, the largest volcano in the Solar System.

Jupiter GAS GIANT

Moons: ~100 confirmed. Key ones — Io, Europa, Ganymede (the Solar System's largest moon, bigger than Mercury), Callisto. Famous for its Great Red Spot storm.

Saturn GAS GIANT

Moons: ~280+ confirmed (the count keeps rising as tiny new ones are found). Key ones — Titan (2nd-largest moon, thick atmosphere), Enceladus (icy geysers). Famous for its ring system.

Uranus ICE GIANT

Moons: 28. Key ones — Titania, Oberon, Miranda, Ariel, Umbriel. Rotates almost on its side (~98° axial tilt).

Neptune ICE GIANT

Moons: 14. Key one — Triton (orbits backwards; likely a captured dwarf planet). Has the fastest winds in the Solar System.

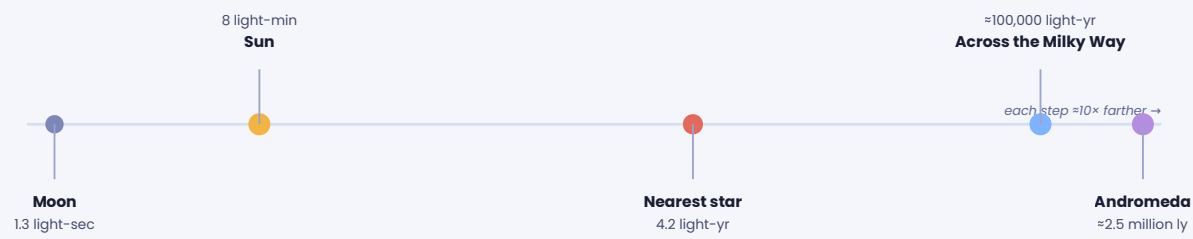
Pluto DWARF PLANET

Moons: 5 — Charon (about half Pluto's size; they orbit almost like a double planet), plus Nix, Hydra, Kerberos, Styx.

03

Distances Worth Remembering

Given in light-time — the time light itself takes to cross the gap — because that's easier to hold onto than a string of zeros.



Log scale: the Moon, Sun and nearest star sit close together compared with the jump to another galaxy.

Moon → Earth	**≈ 1.3 light-seconds**
about 384,000 km — light barely takes over a second.	
Sun → Earth	**≈ 8 light-minutes (1 AU)**
about 150 million km. You always see the Sun as it was 8 minutes ago.	
Edge of the Solar System	**≈ 1–2 light-years**
the outer edge of the Oort Cloud.	
Nearest star (Proxima Centauri)	**≈ 4.2 light-years**
Across the Milky Way	**≈ 100,000 light-years**
tip to tip; some newer estimates put the faint outer disc closer to 150,000–200,000 ly.	
To the Andromeda Galaxy	**≈ 2.5 million light-years**
Edge of the observable universe	**≈ 46 billion light-years**
bigger than the universe's 13.8-billion-year age, because space itself kept expanding while that light travelled.	

04

Our Galaxy & Beyond

A galaxy is a vast, gravity-bound collection of stars, gas, dust and dark matter.

GALAXY SHAPES

- **Spiral** — a flat disc with curving arms, like the Milky Way and Andromeda.
- **Elliptical** — round or oval, mostly older stars, little new star formation.
- **Irregular** — no defined shape, often smaller and shaped by collisions.

KEY GALAXIES

The Milky Way	**≈ 100,000 light-years across**
our home; a barred spiral with 100–400 billion stars. The Sun sits in the Orion Arm, ≈26,000 ly from the galactic centre.	
Andromeda (M31)	**≈ 220,000 light-years across**
the nearest large galaxy, ≈2.5 million ly away, and roughly twice the Milky Way's width. Approaching us — expected to merge in ≈4–5 billion years.	
Triangulum (M33)	**≈ 2.7 million light-years away**
third-largest member of our Local Group of galaxies.	

FACT

The observable universe is estimated to contain somewhere between 100 billion and 2 trillion galaxies.

05

Stars Worth Knowing

Our closest neighbours, the different life stages a star can reach, and the biggest ones ever measured.

NEAREST STARS (BEYOND THE SUN)

Proxima Centauri — **4.24 light-years**
closest star to the Sun; a small red dwarf with at least one known planet.

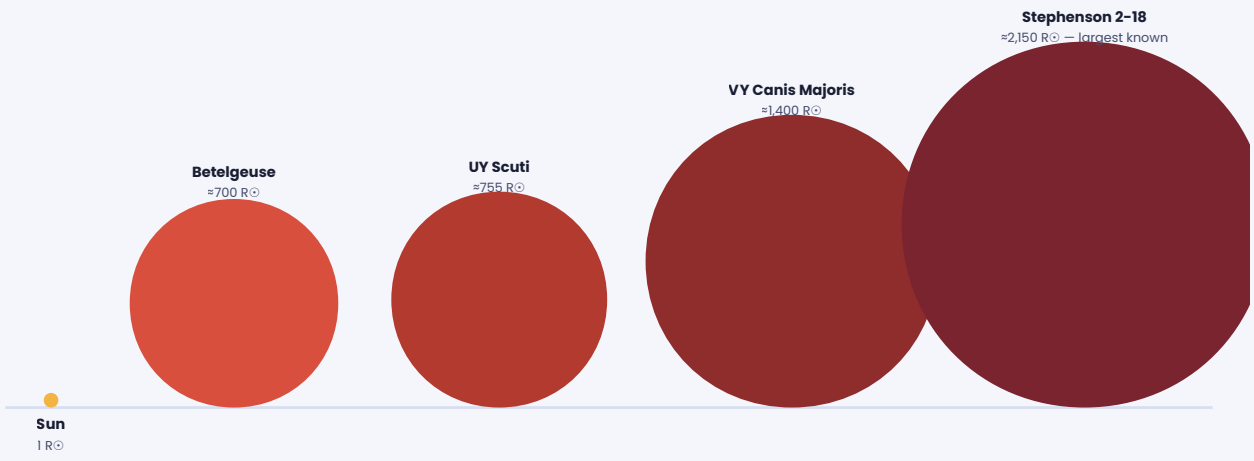
Alpha Centauri A & B — **4.37 light-years**
a Sun-like binary pair, part of the same system as Proxima.

Barnard's Star — **5.96 light-years**
a red dwarf with the fastest apparent motion across our sky.

Sirius — **8.6 light-years**
the brightest star in Earth's night sky.

A STAR'S LIFE STAGES

- **Red dwarf** — smallest, coolest, and by far the most common type of star; can burn for trillions of years.
- **Yellow dwarf** (like our Sun) — medium-sized, roughly a 10-billion-year lifespan.
- **Red giant** — a dying Sun-like star, hugely expanded after using up its core hydrogen.
- **Supergiant** — the largest, most massive stars; short, dramatic lives ending in supernovae.
- **White dwarf** — the hot, dense leftover core of a star like the Sun, after it sheds its outer layers.
- **Neutron star** — the crushed core left after a supergiant explodes; a teaspoon of it would weigh about a billion tonnes.



Star radii compared with the Sun (R☉) — circle sizes are compressed (√-scaled) for display; true ratios are far more extreme.

FOR SCALE

Our Sun's radius is about 696,000 km. Roughly **109 Earths** could line up across its face, and over **1.3 million Earths** could fit inside it by volume.

06

Black Holes

A region of space where gravity is so strong that nothing — not even light — can escape once past its **event horizon**.

Sagittarius A*	**≈ 4.3 million M☉**
the supermassive black hole at our own Milky Way's centre, ≈26,000 ly away. Directly imaged in 2022.	
M87*	**≈ 6.5 billion M☉**
the first black hole ever directly imaged (2019), in the galaxy M87, 55 million ly away.	
TON 618	**≈ 66 billion M☉**
one of the most massive black holes known for years, at the centre of a distant quasar.	
Cosmic Horseshoe black hole	**≈ 36 billion M☉**
announced in 2025; measured with unusually high confidence via gravitational lensing.	
Phoenix A	**≈ 100 billion M☉ (estimated)**
a leading, though less certain, candidate for the most massive black hole known.	
Cygnus X-1	**≈ 21 M☉**
the first black hole ever strongly identified (1964) — a much smaller "stellar-mass" black hole, for comparison.	

NOTE

M☉ means "solar masses" (multiples of the Sun's mass). The very largest black hole masses are estimated indirectly, so rankings can shift as measurements improve.

07

Nebulas

A nebula is a giant cloud of gas and dust — often a nursery where new stars are born, or the remains of a dying one.

Orion Nebula (M42)	**≈ 1,344 light-years away**
visible to the naked eye in Orion's "sword"; one of the closest, brightest stellar nurseries.	
Pillars of Creation	**part of the Eagle Nebula**
towering columns of gas where new stars are forming; made famous by Hubble and JWST images.	
Crab Nebula (M1)	**supernova remnant**
from a star explosion observed and recorded in 1054 AD; holds a pulsar (spinning neutron star) at its centre.	
Ring Nebula (M57)	**planetary nebula**
a glowing shell of gas shed by a dying Sun-like star.	
Helix Nebula	**planetary nebula**
nicknamed the "Eye of God" for its appearance.	

08

Common Star Constellations

Patterns of stars as seen from Earth — the stars themselves are usually nowhere near each other in real space.

Ursa Major — **Great Bear**
contains the "Big Dipper" (or "the Plough") — one of the most recognisable patterns in the northern sky.

Ursa Minor — **Little Bear**
contains the "Little Dipper" and Polaris, the North Star, at the tip of its handle.

Orion — **The Hunter**
spotted easily by its three-star belt; visible from almost everywhere on Earth.

Cassiopeia — **The Queen**
a distinctive "W" or "M" shape in the northern sky.

Scorpius — **The Scorpion**
a large southern-sky constellation with the bright red star Antares at its heart.

Leo — **The Lion**
resembles a crouching lion; contains several bright stars.

Crux — **Southern Cross**
compact and used for navigation in the Southern Hemisphere; not visible from most of the north.

09

The Universe

How old it is, how big we can see, and what it's actually made of.

Age of the universe — **≈ 13.8 billion years**
since the Big Bang.

Observable universe — **≈ 93 billion light-years across**
larger than "twice its age in light-years" because space itself has been expanding the whole time that light was travelling toward us.

What it's made of — **≈5% matter · 27% dark matter · 68% dark energy**
"matter" here includes every star, planet and person ever observed.

FACT

The universe's expansion is **accelerating**, not slowing down — a discovery (1998) that won the 2011 Nobel Prize in Physics, and one still not fully explained.

10

More Interesting Facts

A few things about the universe, stars, supernovae and black holes that are worth knowing simply because they're astonishing.

THE UNIVERSE

The Cosmic Calendar. If the universe's 13.8-billion-year history were squeezed into a single calendar year, all of recorded human history would fit into the last few minutes of December 31st.

Only the visible part. The "observable" universe is just the part whose light has had time to reach us since the Big Bang — the entire universe could be far larger, even infinite.

Racing away forever. Space between distant galaxies is expanding so fast that their light will eventually never reach us at all — the night sky will show fewer galaxies in the far future than it does today.

STARS & THEIR AGES

Middle-aged Sun. The Sun is about 4.6 billion years old and roughly halfway through its 10-billion-year life as a stable, hydrogen-burning star.

Immortal, so far. Red dwarfs burn fuel so slowly that not a single one born since the universe began has died of old age yet.

The first stars. The universe's very first stars formed only 100–200 million years after the Big Bang, from clouds of pure hydrogen and helium — no heavier elements existed yet.

You are stardust. Every atom of carbon, oxygen and iron in your body was forged inside a star that died before the Sun was even born.

SUPERNOVAE

Briefly the brightest thing around. A single supernova can, for a few weeks, outshine its entire host galaxy of billions of stars.

Seen in daylight. The star that created the Crab Nebula exploded in 1054 AD and was recorded by Chinese and Arab astronomers as bright enough to see in broad daylight for weeks.

Where gold comes from. Elements heavier than iron — including gold and silver — are forged in the extreme heat of supernovae and neutron-star collisions.

A star to watch. Betelgeuse, in Orion's shoulder, could go supernova any time in the next 100,000 years — astronomically speaking, "any day now."

BLACK HOLES

Time slows at the edge. Watching someone fall toward a black hole's event horizon, you'd see them appear to freeze there forever — even though they wouldn't feel anything unusual as they crossed it.

Spaghettification. Falling in feet-first, the difference in gravity between your head and feet would stretch you into a thin, noodle-like shape — a real physics term.

Not a vacuum cleaner. If the Sun were suddenly replaced by a black hole of the same mass, Earth's orbit wouldn't change at all — we'd just freeze without sunlight.

They slowly evaporate. Stephen Hawking predicted black holes leak energy over unimaginable timespans (Hawking radiation) — a Sun-sized black hole would still take vastly longer than the universe's current age to fully evaporate.

QUICK QUIZ

Space GK — Test Yourself

Cover the answers and see how many you get before checking.

Q. Which planet is known as the Red Planet? A. Mars.

Q. Which is the largest planet in our Solar System? A. Jupiter.

Q. Which planet has the most spectacular ring system? A. Saturn.

Q. Which planet spins almost on its side? A. Uranus.

Q. What is the closest star to Earth, other than the Sun? A. Proxima Centauri.

Q. What is the name of our home galaxy? A. The Milky Way.

Q. What is the nearest large galaxy to the Milky Way? A. Andromeda (M31).

Q. What do we call a black hole's point of no return? A. The event horizon.

Q. Which black hole was the first ever directly photographed, and when? A. M87*, in 2019.

Q. What does a light-year actually measure? A. Distance — the distance light travels in a year (≈9.46 trillion km) — not time.

Q. What is the farthest human-made object from Earth? A. Voyager 1, launched in 1977.

Q. Why was Pluto reclassified as a dwarf planet in 2006? A. It hadn't cleared its orbital neighbourhood of other debris.

Q. What is the name of the North Star? A. Polaris.

Q. Roughly how old is the universe? A. About 13.8 billion years.

Q. What is a supernova? A. The explosive death of a massive star.

Revisit one section at a time — the distance diagrams are the fastest way to re-anchor the scale of everything else.

Look up, and wonder.